

Problem 1, Cell 2: the orthogonality (chain) lemma

Official statement. A line here always means a line through the origin of $\mathbb{R}^d$. The angle between two lines ℓ, ℓ' is the acute (non-obtuse) angle $\theta(\ell, \ell') \in [0, \pi/2]$ between them. If ℓ, ℓ' are spanned by unit vectors x, x' , then $\theta(\ell, \ell') = \arccos |\langle x, x' \rangle|$. For lines $\ell_1, \dots, \ell_N$ in $\mathbb{R}^d$ (repetitions allowed), write $S(\ell_1, \dots, \ell_N) = \sum_{1 \leq i < j \leq N} \theta(\ell_i, \ell_j)$.

A statement about a chain of vectors, in which each vector may fail to be orthogonal only to its immediate neighbours in the list. Let $m \geq 2$ and let $x_1, \dots, x_m$ be unit vectors in $\mathbb{R}^{m-1}$ with $\langle x_i, x_j \rangle = 0$ whenever $|i - j| \geq 2$. Prove that $\sum_{i=1}^{m-1} \theta(x_i, x_{i+1}) \leq (m - 2) \cdot \pi/2$, where $\theta(x, y) = \arccos |\langle x, y \rangle|$.

Theorem 1 below is exactly this statement: for unit vectors, $\angle(x, y) = \theta(x, y)$.

Role in Problem 1. This lemma is the chain case of the bound for $N = d + 1$ lines in $\mathbb{R}^d$ (the next cell of Problem 1). There, $d + 1$ unit vectors in $\mathbb{R}^d$ are always linearly dependent, and the lemma treats the dependent configurations whose non-orthogonal pairs form a path. The cell asks for exactly this lemma.

Source. The proof below is our own; we found no source for this lemma.

For nonzero $x, y \in \mathbb{R}^n$, the angle between the lines $\mathbb{R}x$ and $\mathbb{R}y$ is

$$\angle(x, y) = \arccos \frac{|\langle x, y \rangle|}{\|x\| \|y\|} \in [0, \pi/2].$$

Theorem 1 (Orthogonality lemma). *Let $m \geq 2$ and let $x_1, \dots, x_m \in \mathbb{R}^{m-1}$ be unit vectors with*

$$\langle x_i, x_j \rangle = 0 \quad \text{whenever } |i - j| \geq 2.$$

Put $\theta_i = \angle(x_i, x_{i+1})$ for $1 \leq i \leq m - 1$. Then

$$\sum_{i=1}^{m-1} \theta_i \leq \frac{(m - 2)\pi}{2}.$$

Remark 2. Only $\dim \text{span}(x_1, \dots, x_m) \leq m - 1$ is used. So the theorem also holds for unit vectors in any $\mathbb{R}^n$ that span a subspace of dimension at most $m - 1$.

1 Reductions

Sign normalisation. Replacing some x_i by $-x_i$ changes neither the lines $\mathbb{R}x_i$ (so neither the θ_i) nor the orthogonality hypotheses nor the ambient space. We choose signs one index at a time. Keep x_1 . For $i = 1, 2, \dots, m - 1$ in turn, replace x_{i+1} by $-x_{i+1}$ if $\langle x_i, x_{i+1} \rangle < 0$. Step i changes only x_{i+1} . So it does not affect $\langle x_{i'}, x_{i'+1} \rangle$ for $i' < i$, whose vectors have indices at most i and are already fixed. After step $m - 1$,

$$c_i := \langle x_i, x_{i+1} \rangle \geq 0 \quad (1 \leq i \leq m - 1).$$

Since $\|x_i\| = \|x_{i+1}\| = 1$, we get $c_i \in [0, 1]$ and $\theta_i = \arccos c_i$.

Complementary angles. Put $\alpha_i = \frac{\pi}{2} - \theta_i \in [0, \frac{\pi}{2}]$. Then $c_i = \cos \theta_i = \sin \alpha_i$. Since

$$\sum_{i=1}^{m-1} \theta_i = \frac{(m-1)\pi}{2} - \sum_{i=1}^{m-1} \alpha_i,$$

the theorem is equivalent to

$$\sum_{i=1}^{m-1} \alpha_i \geq \frac{\pi}{2}. \quad (1)$$

2 Gram–Schmidt residuals

Let $V_0 = \{0\}$ and $V_j = \text{span}(x_1, \dots, x_j)$ for $1 \leq j \leq m$. Let P_j be the orthogonal projection onto V_j (so $P_0 = 0$). Define

$$u_j = x_j - P_{j-1}x_j, \quad r_j = \|u_j\|^2 = \text{dist}(x_j, V_{j-1})^2 \quad (1 \leq j \leq m).$$

Thus $u_j \perp V_{j-1}$ and, by Pythagoras, $r_j = 1 - \|P_{j-1}x_j\|^2$. Also $r_1 = \|x_1\|^2 = 1$. Finally, $r_j > 0$ if and only if $x_j \notin V_{j-1}$.

For $1 \leq j \leq m$ set $\beta_j = \alpha_1 + \dots + \alpha_{j-1}$ (so $\beta_1 = 0$). Since every $\alpha_i \geq 0$, the sequence β_j is nondecreasing.

Lemma 3 (Recursion). *Let $1 \leq j \leq m-1$ and suppose $r_j > 0$. Then $r_{j+1} = 1 - \frac{c_j^2}{r_j}$.*

Proof. Since $u_j \in V_j$, $u_j \perp V_{j-1}$, and $x_j = P_{j-1}x_j + u_j$, we have the orthogonal decomposition $V_j = V_{j-1} \oplus \mathbb{R}u_j$. Here $u_j \neq 0$ because $r_j > 0$. Hence

$$P_j y = P_{j-1}y + \frac{\langle y, u_j \rangle}{r_j} u_j \quad (y \in \mathbb{R}^{m-1}).$$

Apply this to $y = x_{j+1}$. By hypothesis $x_{j+1} \perp x_1, \dots, x_{j-1}$ (their indices differ from $j+1$ by at least 2), so $x_{j+1} \perp V_{j-1}$. (For $j=1$ this is trivial, as $V_0 = \{0\}$.) Therefore $P_{j-1}x_{j+1} = 0$. Moreover, since $P_{j-1}x_j \in V_{j-1}$,

$$\langle x_{j+1}, u_j \rangle = \langle x_{j+1}, x_j \rangle - \langle x_{j+1}, P_{j-1}x_j \rangle = c_j - 0 = c_j.$$

Hence $P_j x_{j+1} = \frac{c_j}{r_j} u_j$ and $\|P_j x_{j+1}\|^2 = \frac{c_j^2}{r_j^2} r_j = \frac{c_j^2}{r_j}$. Then $r_{j+1} = 1 - \|P_j x_{j+1}\|^2 = 1 - c_j^2/r_j$. $\square$

Lemma 4 (Trigonometric inequality). *Let $a, b \in [0, \pi/2]$ with $a+b < \pi/2$. Then*

$$1 - \frac{\sin^2 a}{\cos^2 b} \geq \cos^2(a+b).$$

Proof. Note $\cos b > 0$ because $b \leq a+b < \pi/2$. The identity $\cos(x+y)\cos(x-y) = \cos^2 x - \sin^2 y$ gives

$$\cos(b+a)\cos(b-a) = \cos^2 b \cos^2 a - \sin^2 b \sin^2 a = \cos^2 b(1 - \sin^2 a) - (1 - \cos^2 b) \sin^2 a = \cos^2 b - \sin^2 a.$$

Also $\cos(b-a) - \cos(b+a) = 2 \sin a \sin b \geq 0$ because $a, b \in [0, \pi/2]$. Now $\cos(a+b) > 0$, hence also $\cos(b-a) \geq \cos(a+b) > 0$, and $0 < \cos^2 b \leq 1$. Hence

$$1 - \frac{\sin^2 a}{\cos^2 b} = \frac{\cos(a+b)\cos(b-a)}{\cos^2 b} \geq \cos(a+b)\cos(b-a) \geq \cos(a+b)\cos(a+b). \quad \square$$

Claim 5. For every $j \in \{1, \dots, m\}$ with $\beta_j < \pi/2$ we have $r_j \geq \cos^2 \beta_j > 0$.

Proof. Induction on j . For $j = 1$: $\beta_1 = 0$ and $r_1 = 1 = \cos^2 0$.

Step $j \rightarrow j+1$ ($1 \leq j \leq m-1$). Assume $\beta_{j+1} = \beta_j + \alpha_j < \pi/2$. Then $\beta_j \leq \beta_{j+1} < \pi/2$, so by the induction hypothesis $r_j \geq \cos^2 \beta_j > 0$. By Lemma 3, $r_{j+1} = 1 - c_j^2/r_j$. Using $c_j^2 = \sin^2 \alpha_j$ and $r_j \geq \cos^2 \beta_j > 0$, we get $c_j^2/r_j \leq \sin^2 \alpha_j / \cos^2 \beta_j$. Hence, by Lemma 4 with $a = \alpha_j$ and $b = \beta_j$ (both lie in $[0, \pi/2]$, and $a + b = \beta_{j+1} < \pi/2$),

$$r_{j+1} \geq 1 - \frac{\sin^2 \alpha_j}{\cos^2 \beta_j} \geq \cos^2(\alpha_j + \beta_j) = \cos^2 \beta_{j+1} > 0,$$

where the last inequality holds because $0 \leq \beta_{j+1} < \pi/2$. $\square$

3 Proof of Theorem 1

Suppose, for contradiction, that (1) fails, i.e. $\beta_m = \sum_{i=1}^{m-1} \alpha_i < \pi/2$. Because (β_j) is nondecreasing, $\beta_j < \pi/2$ for every $j = 1, \dots, m$. By Claim 5, $r_j > 0$ for all j , i.e. $x_j \notin \text{span}(x_1, \dots, x_{j-1})$ for all j . Then $x_1, \dots, x_m$ are linearly independent. (If $\sum \lambda_i x_i = 0$ with not all $\lambda_i = 0$, let j be the largest index with $\lambda_j \neq 0$; then $x_j \in V_{j-1}$, a contradiction.) This gives m linearly independent vectors in $\mathbb{R}^{m-1}$, which is impossible. Hence $\sum \alpha_i \geq \pi/2$, which is (1), and so $\sum \theta_i \leq (m-2)\pi/2$. $\square$

4 Edge cases

- $m = 2$: the vectors lie in $\mathbb{R}^1$, so $x_2 = \pm x_1$ and $\theta_1 = 0 = (m-2)\pi/2$. The proof covers this case: $\alpha_1 < \pi/2$ would give $r_2 = 1 - \sin^2 \alpha_1 = \cos^2 \alpha_1 > 0$, i.e. two independent vectors in $\mathbb{R}^1$. There are no orthogonality hypotheses when $m = 2$.
- $c_i = 1$ for some i (i.e. $x_{i+1} = x_i$ after normalisation): then $\alpha_i = \pi/2$ and (1) holds trivially. In the argument, $\beta_j \geq \pi/2$ for $j > i$, so the claim is never invoked beyond index i . In particular we never divide by an r_j that might vanish.
- $c_i = 0$: then $\alpha_i = 0$, and the recursion gives $r_{i+1} = 1$, consistent with $x_{i+1} \perp V_i$.
- Division by r_j occurs only in Lemma 3, and only under the hypothesis $r_j > 0$. In Claim 5 that hypothesis comes from the induction.

Remark 6 (Sharpness). Equality is attained. For example, take $x_1 = e_1, x_2 = e_2, \dots, x_{m-1} = e_{m-1}$ and $x_m = e_{m-1}$ in $\mathbb{R}^{m-1}$. Then $\theta_1 = \dots = \theta_{m-2} = \pi/2$ and $\theta_{m-1} = 0$. For $m = 3$, the chain $e_1, (\cos t, \sin t), e_2$ in $\mathbb{R}^2$ with $t \in [0, \pi/2]$ gives equality for every t .